

# Module 4 Exercises: Computational Thinking

## Computational Thinking with GenAI:

### LEARNER WORKBOOK

For this lesson, all the code you need is here or comes from Claude.

#### How to use this file:

- Run ONE section at a time. Highlight the lines, then press Ctrl+Enter (Windows) or Cmd+Enter (Mac).
- Paste code from Claude where it says PASTE HERE.
- If code gives an error, copy the full error message to Claude.
- Put chem\_data.csv in the same folder as this file, then use  
Session > Set Working Directory > To Source File Location.

```
library(tidyverse)
chem <- read_csv("../data/chem_data.csv")
```

#### Scenario:

#### ICU clinicians say their chemistry results come back slower

As the lab director, you've requested data to help you determine

- Q1. Does turnaround time (TAT) differ by patient type?
- Q2. Are we meeting a 3-hour collect-to-verify goal?

```
# RUN THIS CODE CHUNK TO LOAD REQUIRED PACKAGES
library(tidyverse)
```

## Build your data subset

Fill in the two blanks with “potassium” and “creatinine”. Keep the quote marks. This will create a data set that only includes data from glucose, potassium, and creatinine tests.

```
tests_keep <- c("glucose", , )

read_csv("chem_data.csv", show_col_types = FALSE) |>
  filter(test %in% tests_keep) |>
  write_csv("chem_subset.csv")

chem <- read_csv("chem_subset.csv", show_col_types = FALSE)

glimpse(chem)
```

CHECK: you should have 19,443 rows (6,481 patients x 3 tests).

## INITIAL GEN AI PROMPT

Do not upload the data file to Claude.

Paste your prompt here:

Paste the code Claude gave you in the code chunk below. Run it. What happened?

```
# PASTE CODE FROM CLAUDE HERE AND RUN IT
```

## EXERCISE 1. Decomposition (Objective 1)

No code. Follow the slides. You can type your response below.

## EXERCISE 2. Pattern recognition (Objective 2)

The code below works for ONE test (glucose) in ONE patient type (icu). Run it first.

```
chem |> filter(test == "glucose", pat_type == "icu") |>
  summarize(n_results = n(), median_value = median(value))
```

### A) Claude's version for every patient type, glucose only

```
# PASTE CODE FROM CLAUDE HERE AND RUN IT
```

### B) Claude's version for every test and every patient type

```
# PASTE CODE FROM CLAUDE HERE AND RUN IT
```

## EXERCISE 3. Abstraction (Objective 3)

Run both code chunks below. Then inspect the code in each chunk to find what differs between them.

```
chem |> filter(test == "glucose") |>
  group_by(pat_type) |>
  summarize(n_results = n(), pct_above = round(100 * mean(value > 125), 1))
```

Answer:

```
chem |> filter(test == "potassium") |>
  group_by(pat_type) |>
  summarize(n_results = n(), pct_above = round(100 * mean(value > 5.0), 1))
```

Answer:

What differs between the two code chunks?

**Claude's code from your filled-in prompt template**

```
# PASTE CODE FROM CLAUDE HERE AND RUN IT
```

## EXERCISE 4. Algorithm design (Objective 4)

### REVISED GEN AI PROMPT

Paste your prompt here:

**Paste the code Claude wrote AFTER you answered its questions and approved its plan.**

```
# PASTE CODE FROM CLAUDE HERE AND RUN IT
```

CHECK: the patient counts in your table should total 6,481, and patients with no patient type should still be there as a group.

### DEBRIEF. Does this make sense?

```
n_distinct(chem$mrn) # how many patients are in the data? Do they total 6,481?  
nrow(chem)/n_distinct(chem$mrn) # how many rows per patient? Is there 3 (one for each test)
```

If the total number of patients is 5,393, patients with no patient type were dropped.